# MDX File Format Specification

**Format**: Blizzard MDX Model Format (Warcraft III / Warcraft III: Reforged) **Byte Order**: Little-endian **File Magic**: `MDLX` **Supported Versions**: 800 (Classic), 900–1200 (Reforged)

This document describes the MDX format used by Warcraft III (Classic and Reforged). The format is chunk-based and supports skeletal animation, particle effects, cameras, and collision volumes.

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## Revision History

| Date | Version | Description |
|---|---|---|
| 2025-07-18 | 2.0 | Complete rewrite based on WhiteoutLib parser/writer implementation, GhostWolf's HiveWorkshop reference, and validation against 5081-file Reforged corpus |
| 2026-05-21 | 2.1 | Corrected the Layer fresnel-field version gate (v1000, not v900); documented the v≥1100 layer `shader` field as a `ShaderType` enum (was mislabelled `isHD`); fixed the KEVT field order (`globalSequenceId` precedes the track times — v2.0 had it backwards); added field defaults, signed keyframe-time notes, and the precise version→release mapping. Validated by a byte-exact MDX→MDL→MDX round-trip over the full `war3.w3mod` game-asset corpus. |

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# 1. Overview

MDX is a **binary 3D model format** created by Blizzard Entertainment for Warcraft III and Warcraft III: Reforged. It stores geometry, materials, skeletal animation, particle effects, cameras, collision volumes, and other data needed to render and animate game models.

The format is **chunk-based**: the file consists of a 4-byte magic number followed by a sequence of tagged chunks in no prescribed order. All chunks are optional. All multi-byte values are **little-endian**.

## Known Versions

| Version | Game |
|---|---|
| 800 | Warcraft III: Reign of Chaos / The Frozen Throne (Classic) |
| 900 | Warcraft III: Reforged — Beta |
| 1000 | Warcraft III: Reforged — First Release (1.32) |
| 1100 | Warcraft III: Reforged — 2.0.0 and later |
| 1200 | Warcraft III: Reforged — current 2.0.x |

> **Corpus note:** A corpus of 5081 MDX files extracted from Warcraft III: Reforged consists entirely of version 1200 files. Classic v800 models exist in older game data.

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# 2. Conventions

## Data Types

| Notation | Size | Description |
|---|---|---|
| `u8` | 1 | Unsigned 8-bit integer |
| `u16` | 2 | Unsigned 16-bit integer |
| `u32` | 4 | Unsigned 32-bit integer |
| `i32` | 4 | Signed 32-bit integer |
| `f32` | 4 | IEEE 754 single-precision float |
| `char[N]` | N | Fixed-length null-padded ASCII string |
| `float[N]` | 4×N | Array of N `f32` values |
| `FourCC` | 4 | Four ASCII characters read as a `u32` |

## Notation

- `(X)` means field or sub-chunk X is **optional** and may or may not be present.
- `[N]` after a type means an array of N elements.
- `[?]` means the array count is not stored directly; see parsing notes.
- `// version > V` means the field only exists when the model version exceeds V.
- Flag fields use hexadecimal bit masks (e.g., `0x1`, `0x2`).

## FourCC Tags

Chunk identifiers are 4-byte ASCII strings stored as little-endian `u32` values. For example, `"MDLX"` is the bytes `0x4D`, `0x44`, `0x4C`, `0x58`, which as a little-endian `u32` is `0x584C444D`. In code, define constants like:

```
constexpr u32 makeTag(char a, char b, char c, char d) {
    return static_cast<u32>(a) | (static_cast<u32>(b) << 8) |
           (static_cast<u32>(c) << 16) | (static_cast<u32>(d) << 24);
}
constexpr u32 MDLX_TAG = makeTag('M', 'D', 'L', 'X');
```

---

## 3. File Structure

```
MDX {
    FourCC  "MDLX"          // Magic number (0x584C444D)
    (VERS)                   // Version
    (MODL)                   // Model info
    (SEQS)                   // Animation sequences
    (GLBS)                   // Global sequences
    (TEXS)                   // Textures
    (SNDS)                   // Sounds (deprecated, unused in shipped
models)
    (SNEM)                   // Sound emitters (deprecated, unused in
shipped models)
    (MTLS)                   // Materials
    (TXAN)                   // Texture animations
    (GEOS)                   // Geosets (mesh data)
    (GEOA)                   // Geoset animations
    (BONE)                   // Bones
    (LITE)                   // Lights
    (HELP)                   // Helpers
    (ATCH)                   // Attachments
    (PIVT)                   // Pivot points
    (PREM)                   // Particle emitters (model-based)
    (PRE2)                   // Particle emitters 2 (quad-based)
    (RIBB)                   // Ribbon emitters
    (EVTS)                   // Event objects
    (CAMS)                   // Cameras
    (CLID)                   // Collision shapes
    (BPOS)                   // Bind poses (version > 800)
    (FAFX)                   // Face effects (version > 800)
    (CORN)                   // Popcorn emitters (version > 800)
}
```

All chunks are optional and may appear in any order. A valid MDX file can consist of just the 4-byte `"MDLX"` magic. Unknown chunks should be skipped using the size from the chunk header.

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## 4. Chunk Header

Every top-level chunk (and many sub-chunks) begins with an 8-byte header:

```
ChunkHeader {
    FourCC  tag             // 4-byte ASCII identifier
    u32     size            // Size in bytes of chunk data (excludes this
8-byte header)
}
```

To skip an unknown chunk, read the header and advance `size` bytes.

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# 5. Parsing Strategy

```
read magic "MDLX" (4 bytes)

while (bytesRemaining > 0) {
    tag  = readU32()    // FourCC
    size = readU32()    // Chunk data size
    endPos = currentPos + size

    switch (tag) {
        case VERS_TAG: parseVERS(size); break;
        case MODL_TAG: parseMODL(size); break;
        // ... handle known chunks ...
        default: skip(size); break;     // Unknown chunk
    }

    assert(currentPos == endPos)        // Validate exact consumption
}
```

## Variable-Sized Object Chunks

Many chunks contain arrays of variable-sized objects. Two patterns exist:

**Pattern 1 — Objects with `inclusiveSize`:** Materials, layers, geosets, geoset animations, lights, attachments, particle emitters, particle emitter 2s, ribbon emitters, cameras, texture animations, and corn emitters store an `inclusiveSize` at the start of each object. This size includes the 4 bytes of the `inclusiveSize` field itself.

```
totalRead = 0
while (totalRead < chunkSize) {
    inclusiveSize = readU32()
    parseObject(inclusiveSize - 4)      // Remaining bytes
    totalRead += inclusiveSize
}
```

**Pattern 2 — Objects without `inclusiveSize`:** Bones, event objects, helpers, and collision shapes do **not** have an explicit `inclusiveSize`. Their size must be computed:

- **Bone**: Node `inclusiveSize` + 8 bytes (geosetId + geosetAnimationId)
- **Helper**: Node `inclusiveSize` exactly
- **Event Object**: Node `inclusiveSize` + KEVT sub-chunk size (if present)
- **Collision Shape**: Node `inclusiveSize` + type-dependent vertex data + optional radius

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# 6. Common Structures

## 6.1 Extent

Size: **28 bytes**

```
Extent {
    f32         boundsRadius
    float[3]    minimum             // Bounding box min (x, y, z)
    float[3]    maximum             // Bounding box max (x, y, z)
}
```

## 6.2 Node

See Section 9 — Node Structure.

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# 7. Top-Level Chunks

## 7.1 VERS — Version

**Tag:** `0x53524556` · **Fixed size:** 4 bytes

```
VERS {
    u32     version                 // 800, 900, 1000, 1100, or 1200
}
```

## 7.2 MODL — Model Info

**Tag:** `0x4C444F4D` · **Fixed size:** 372 bytes

```
MODL {
    char[80]    name                // Model name
    char[260]   animationFileName   // Animation file path
    Extent      extent              // Model bounds (28 bytes)
    u32         blendTime           // Animation blend time (ms)
}
```

## 7.3 SEQS — Sequences

**Tag:** `0x53514553` · **Element size:** 132 bytes · **Count:** `size / 132`

```
SEQS {
    Sequence[size / 132] sequences
}

Sequence {                          // 132 bytes
    char[80]    name
    u32         intervalStart       // Start time (ms)
    u32         intervalEnd         // End time (ms)
    f32         moveSpeed
    u32         flags               // 0 = looping, 1 = non-looping
    f32         rarity              // Playback probability weight
    u32         syncPoint
    Extent      extent              // 28 bytes
}
```

## 7.4 GLBS — Global Sequences

**Tag:** `0x53424C47` · **Element size:** 4 bytes · **Count:** `size / 4`

```
GLBS {
    u32[size / 4]   durations       // Duration of each global sequence
(ms)
}
```

### 7.5 TEXS — Textures

**Tag:** `0x53584554` · **Element size:** 268 bytes · **Count:** `size / 268`

```
TEXS {
    Texture[size / 268] textures
}

Texture {                           // 268 bytes
    u32         replaceableId       // 0 = none, 1 = team color, 2 = team
glow, etc.
    char[260]   fileName            // Texture file path (.blp / .tga)
    u32         flags               // 0x1 = wrap width, 0x2 = wrap height
}
```

### 7.6 SNDS — Sounds

**Tag:** `0x53444E53` · **Element size:** 56 bytes · **Count:** `size / 56`

> **Note:** This chunk was used before Warcraft III released and is not present in any known shipped model. Readers for it still exist in the internal game code. Documented for completeness based on disassembly research.

```
SNDS {
    Sound[size / 56] sounds
}

Sound {                             // 56 bytes
    char[44]    soundFile           // Path to sound file (null-
terminated)
    f32         maximumDistance     // Maximum audible distance
    f32         minimumDistance     // Minimum distance (full volume)
    u32         soundChannel        // Sound channel identifier
}
```

### 7.6b SNEM — Sound Emitters

**Tag:** `0x4D454E53` · **Variable-sized objects**

> **Note:** Like SNDS, this chunk is not present in any known shipped model but readers exist in the internal game code. Sound emitters are node-based objects that can emit sounds at specific animation frames via KSEK tracks.

```
SNEM {
    SoundEmitter[?] soundEmitters   // Parse using inclusiveSize
}

SoundEmitter {
    u32         inclusiveSize       // Total size including this field
    Node        node                // Base node data with transform
    (KSEK)                          // Sound track: u32 values
}
```

## 7.7 MTLS — Materials

**Tag:** `0x534C544D` · **Variable-sized objects**

```
MTLS {
    Material[?] materials           // Parse using inclusiveSize
}
```

**Material**

```
Material {
    u32         inclusiveSize       // Total size including this field
    u32         priorityPlane
    u32         flags

    // version > 800 AND version < 1100 only:
    char[80]    shader              // Shader name (e.g.
"Shader_HD_DefaultUnit")

    FourCC      "LAYS"              // Layer sub-chunk tag (0x5359414C)
    u32         layerCount
    Layer[layerCount] layers
}
```

> **Important version gate:** The `shader` field exists only when `version > 800 && version < 1100`. In version ≥ 1100, the shader field was removed and the layer system was redesigned with the SubTexture system (see Layer below).

### Layer

```
Layer {
    u32         inclusiveSize       // Total size including this field
    u32         filterMode          // See FilterMode enum
    u32         shadingFlags        // See ShadingFlags bitfield
    u32         textureId           // Index into TEXS
    u32         textureAnimationId  // Index into TXAN, or 0xFFFFFFFF (-1)
    u32         coordId             // UV set index in geoset
    f32         alpha               // Layer opacity (0.0–1.0)

    // version > 800:
    f32         emissiveGain        // Emissive intensity; default 1.0
(not 0.0)

    // version > 900:
    float[3]    fresnelColor        // RGB; default (1, 1, 1)
    f32         fresnelOpacity
    f32         fresnelTeamColor

    // version >= 1100 (SubTexture system):
    u32         shader              // ShaderType enum (0 = SD, 1 = HD, …)
    u32         numSubTextures
    SubTexture[numSubTextures] subTextures

    // Animation tracks:
    (KMTF)                          // version < 1100 only
    (KMTA)
    (KMTE)                          // version > 800
    (KFC3)                          // version > 900
    (KFCA)                          // version > 900
    (KFTC)                          // version > 900
}
```

> **Version gate — fresnel fields:** `emissiveGain` exists when `version > 800`, but `fresnelColor`, `fresnelOpacity`, and `fresnelTeamColor` exist only when `version > 900`. Gating all four on `version > 800` (a common mistake) makes a v900 layer consume 16 bytes of the following track chunk as fresnel data, yielding NaN fresnel values and a desynchronised track stream. The fresnel *track* chunks (KFC3/KFCA/KFTC) share the same `version > 900` gate.

> `shader` **field (version ≥ 1100):** A `u32` holding a `ShaderType` enum value — **not** a boolean `isHD`. SD layers use `0`, HD (PBR) layers use `1`; other values exist for engine-internal post-process shaders (see ShaderType enum).

### SubTexture (version ≥ 1100)

```
SubTexture {
    u32         textureId           // Index into TEXS
    u32         slot                // Texture slot (see SubTextureSlot
enum)
    (KMTF)                          // Per-subtexture texture ID animation
}
```

When `version >= 1100`, the KMTF track is stored per-SubTexture rather than on the Layer itself.

### SubTextureSlot Enum

HD layers in Reforged use a PBR (Physically Based Rendering) texture pipeline. Each slot identifies the role of the texture in the shading model. An HD layer (`shader == 1`) typically has 6 SubTextures — one per slot — while SD layers (`shader == 0`) have a single SubTexture at slot 0.

| Value | Name | Description |
|---|---|---|
| 0 | Diffuse | Albedo / base color map. In SD layers this is the only texture. |
| 1 | Normal | Tangent-space normal map. Default: `Textures/normal.blp` (flat normal). |
| 2 | ORM | Packed Occlusion (R), Roughness (G), Metallic (B) map. Default: `Textures/ORM.blp`. |
| 3 | Emissive | Emissive / self-illumination map. Default: `Textures/Black32.blp` (no emission). |
| 4 | Team Color | Replaceable team color texture. Typically references a texture entry with an empty filename and `replaceableId = 1`. |
| 5 | Environment Map | Cube/sphere environment reflection map. Always `ReplaceableTextures/EnvironmentMap.blp` in the corpus. |

> **Corpus validation (5081 v1200 files, 86321 SubTextures):**
>
> - **14,076 layers** use the full 6-slot HD set `{0, 1, 2, 3, 4, 5}`.
> - **1,349 layers** use a single slot 0 (SD diffuse-only).
> - **82 layers** use `{0, 2, 3, 4, 5}` (HD without normal map).
> - Slots 1–5 never appear without slot 0. Slot values outside 0–5 were not observed.

### ShaderType Enum (Layer `shader`, version ≥ 1100)

The `u32 shader` field selects the shading pipeline for the layer. Models on disk use only `0` (SD) and `1` (HD); the remaining values name engine-internal shaders (post-processing, terrain, UI, …) and are listed for completeness.

| Value | Name | Notes |
|---|---|---|
| 0 | SD | Classic fixed-function shading; single diffuse SubTexture at slot 0. |
| 1 | HD | Reforged PBR shading; the 6-slot SubTexture set. |
| 2 | SDOnHD | SD content rendered through the HD pipeline. |
| 3–25 | *(engine-internal)* | Terrain, Water, Fog, Foliage, Sprite, post-process (DepthOfField, Bloom*, GaussianBlur, Tonemap, CMAA*), Distortion, Crystal, Imgui, … — not used by on-disk model layers. |

**FilterMode Enum**

| Value | Name | Description |
|---|---|---|
| 0 | None | Opaque |
| 1 | Transparent | Alpha-tested (threshold ~0.75) |
| 2 | Blend | Standard alpha blending |
| 3 | Additive | Additive blending (src alpha × src + dst) |
| 4 | AddAlpha | Add alpha (same as additive in practice) |
| 5 | Modulate | Multiply (zero × dst + src color × dst) |
| 6 | Modulate2x | Modulate 2× |

**ShadingFlags Bitfield**

| Bit | Hex | Description |
|---|---|---|
| 0 | 0x1 | Unshaded |
| 1 | 0x2 | Sphere environment map |
| 2 | 0x4 | Wrap width (unknown) |
| 3 | 0x8 | Wrap height (unknown) |
| 4 | 0x10 | Two sided |
| 5 | 0x20 | Unfogged |
| 6 | 0x40 | No depth test |
| 7 | 0x80 | No depth set |
| 8 | 0x100 | Unlit (version > 800) |

### 7.8 TXAN — Texture Animations

**Tag:** `0x4E415854` · **Variable-sized objects**

```
TXAN {
    TextureAnimation[?] animations  // Parse using inclusiveSize
}

TextureAnimation {
    u32         inclusiveSize
    (KTAT)                          // Translation: float[3]
    (KTAR)                          // Rotation: float[4] (quaternion
XYZW)
    (KTAS)                          // Scaling: float[3]
}
```

> **Note:** KTAR uses a **quaternion** `float[4]` rotation, matching GhostWolf's original specification.

### 7.9 GEOS — Geosets

**Tag:** `0x534F4547` · **Variable-sized objects**

```
GEOS {
    Geoset[?] geosets               // Parse using inclusiveSize
}
```

### Geoset

```
Geoset {
    u32         inclusiveSize

    // Vertex positions
    FourCC      "VRTX"              // 0x58545256
    u32         vertexCount
    float[vertexCount * 3] positions

    // Vertex normals
    FourCC      "NRMS"              // 0x534D524E
    u32         normalCount
    float[normalCount * 3] normals

    // Primitive type groups
    FourCC      "PTYP"              // 0x50595450
    u32         faceTypeGroupCount
    u32[faceTypeGroupCount] faceTypeGroups  // Usually 4 (triangles)

    // Primitive count groups
    FourCC      "PCNT"              // 0x544E4350
    u32         faceGroupCount
    u32[faceGroupCount] faceGroups

    // Face indices
    FourCC      "PVTX"              // 0x58545650
    u32         faceCount
    u16[faceCount] faces

    // Vertex group assignments
    FourCC      "GNDX"              // 0x58444E47
    u32         vertexGroupCount
    u8[vertexGroupCount] vertexGroups

    // Matrix group sizes
    FourCC      "MTGC"              // 0x4347544D
    u32         matrixGroupCount
    u32[matrixGroupCount] matrixGroups

    // Matrix indices
    FourCC      "MATS"              // 0x5354414D
    u32         matrixIndexCount
    u32[matrixIndexCount] matrixIndices

    u32         materialId          // Index into MTLS
    u32         selectionGroup
    u32         selectionFlags

    // version > 800:
```

```
    u32         lod                 // Level of detail
    char[80]    lodName             // LOD name

    Extent      extent              // Geoset bounds
    u32         sequenceExtentCount
    Extent[sequenceExtentCount] sequenceExtents

    // version > 800 (optional sub-chunks):
    (Tangents)
    (Skin)

    // Texture coordinate sets
    FourCC      "UVAS"              // 0x53415655
    u32         uvSetCount
    TextureCoordinateSet[uvSetCount] uvSets
}
```

**Tangents Sub-Chunk (version > 800)**

```
Tangents {
    FourCC      "TANG"              // 0x474E4154
    u32         count
    float[count * 4] tangents       // x, y, z, handedness (w)
}
```

**Skin Sub-Chunk (version > 800)**

```
Skin {
    FourCC      "SKIN"              // 0x4E494B53
    u32         count               // Byte count
    u8[count]   data                // Groups of 8 bytes: 4 bone indices +
4 weights
}
```

The SKIN data replaces the GNDX/MTGC/MATS skinning system. Each vertex uses 8 bytes: 4 `u8` bone indices followed by 4 `u8` weights (divide by 255.0 to normalize; weights should sum to 1.0).

**TextureCoordinateSet**

```
TextureCoordinateSet {
    FourCC      "UVBS"              // 0x53425655
    u32         count
    float[count * 2] coordinates    // u, v pairs
}
```

**Primitive Types (faceTypeGroups values)**

| Value | Type |
|---|---|
| 0 | Points |
| 1 | Lines |
| 2 | Line loop |
| 3 | Line strip |
| 4 | Triangles |
| 5 | Triangle strip |
| 6 | Triangle fan |
| 7 | Quads |
| 8 | Quad strip |
| 9 | Polygons |

In practice, virtually all models use type 4 (triangles).

## 7.10 GEOA — Geoset Animations

**Tag:** `0x414F4547` · **Variable-sized objects**

```
GEOA {
    GeosetAnimation[?] animations  // Parse using inclusiveSize
}

GeosetAnimation {
    u32         inclusiveSize
    f32         alpha               // Static alpha value
    u32         flags               // 0x1 = color animation
    float[3]    color               // Static color (BGR order)
    u32         geosetId            // Index of affected geoset

    (KGAO)                          // Alpha animation: f32
    (KGAC)                          // Color animation: float[3]
}
```

## 7.11 BONE — Bones

**Tag:** `0x454E4F42` · **No inclusiveSize** — compute from Node

```
BONE {
    Bone[?] bones
}

Bone {
    Node        node
    u32         geosetId            // Associated geoset, or 0xFFFFFFFF
    u32         geosetAnimationId   // Associated geoset animation, or
0xFFFFFFFF
}
```

**Size computation:** Each Bone is `node.inclusiveSize + 8` bytes.

## 7.12 LITE — Lights

**Tag:** `0x4554494C` · **Variable-sized objects**

```
LITE {
    Light[?] lights                // Parse using inclusiveSize
}

Light {
    u32         inclusiveSize
    Node        node
    u32         type               // See LightType enum
    f32         attenuationStart
    f32         attenuationEnd
    float[3]    color              // RGB
    f32         intensity
    float[3]    ambientColor       // RGB
    f32         ambientIntensity

    // version >= 1200:
    f32         shadowIntensity

    (KLAS)                         // Attenuation start: f32
    (KLAE)                         // Attenuation end: f32
    (KLAC)                         // Color: float[3]
    (KLAI)                         // Intensity: f32
    (KLBI)                         // Ambient intensity: f32
    (KLBC)                         // Ambient color: float[3]
    (KLAV)                         // Visibility: f32
}
```

**LightType Enum**

| Value | Type |
|---|---|
| 0 | Omni |
| 1 | Directional |
| 2 | Ambient |

## 7.13 HELP — Helpers

**Tag:** `0x504C4548` · **No inclusiveSize**

```
HELP {
    Helper[?] helpers
}

Helper {
    Node        node
}
```

**Size computation:** Each Helper is exactly `node.inclusiveSize` bytes.

### 7.14 ATCH — Attachments

**Tag:** `0x48435441` · **Variable-sized objects**

```
ATCH {
    Attachment[?] attachments       // Parse using inclusiveSize
}

Attachment {
    u32         inclusiveSize
    Node        node
    char[260]   path                // Attachment model path
    u32         attachmentId

    (KATV)                          // Visibility: f32
}
```

### 7.15 PIVT — Pivot Points

**Tag:** `0x54564950` · **Element size:** 12 bytes · **Count:** `size / 12`

```
PIVT {
    float[size / 4] points          // Groups of 3 floats (x, y, z)
}
```

Each pivot point is 12 bytes (3 × `f32`). There is one pivot point per node (bones, helpers, lights, attachments, emitters, etc.) plus one per camera.

### 7.16 PREM — Particle Emitters (Model)

**Tag:** `0x4D455250` · **Variable-sized objects**

Emitters that spawn model instances.

```
PREM {
    ParticleEmitter[?] emitters     // Parse using inclusiveSize
}

ParticleEmitter {
    u32         inclusiveSize
    Node        node
    f32         emissionRate
    f32         gravity
    f32         longitude
    f32         latitude
    char[260]   spawnModelFileName  // Model to emit
    f32         lifespan
    f32         initialVelocity

    (KPEE)                          // Emission rate: f32
    (KPEG)                          // Gravity: f32
    (KPLN)                          // Longitude: f32
    (KPLT)                          // Latitude: f32
    (KPEL)                          // Lifespan: f32
    (KPES)                          // Speed: f32
    (KPEV)                          // Visibility: f32
}
```

### 7.17 PRE2 — Particle Emitters 2 (Quad)

**Tag:** `0x32455250` · **Variable-sized objects**

Emitters that spawn textured quads (billboards).

```
PRE2 {
    ParticleEmitter2[?] emitters    // Parse using inclusiveSize
}

ParticleEmitter2 {
    u32         inclusiveSize
    Node        node
    f32         speed
    f32         variation
    f32         latitude
    f32         gravity
    f32         lifespan
    f32         emissionRate
    f32         length
    f32         width

    u32         filterMode          // 0=blend, 1=additive, 2=modulate,
3=modulate2x, 4=alphaKey
    u32         rows                // Texture atlas rows
    u32         columns             // Texture atlas columns
    u32         headOrTail          // 0=head, 1=tail, 2=both

    f32         tailLength
    f32         time                // Mid-point time (0.0–1.0)

    float[3]    segmentColor[3]     // 3 segments × RGB (9 floats total)
    u8[3]       segmentAlpha        // 3 alpha values (0–255)
    float[3]    segmentScaling      // 3 scale factors

    u32[3]      headInterval        // Start/mid/end lifecycle intervals
    u32[3]      headDecayInterval
    u32[3]      tailInterval
    u32[3]      tailDecayInterval

    u32         textureId           // Index into TEXS
    u32         squirt              // 1 = burst mode
    u32         priorityPlane
    u32         replaceableId

    (KP2S)                          // Speed: f32
    (KP2R)                          // Variation: f32
    (KP2L)                          // Latitude: f32
    (KP2G)                          // Gravity: f32
    (KP2E)                          // Emission rate: f32
    (KP2N)                          // Length: f32
    (KP2W)                          // Width: f32
    (KP2V)                          // Visibility: f32
}
```

> **Note:** `length` is written before `width`. Some older specifications had these swapped.

### 7.18 RIBB — Ribbon Emitters

**Tag:** `0x42424952` · **Variable-sized objects**

Emitters that produce connected line segments forming ribbon-like quads.

```
RIBB {
    RibbonEmitter[?] emitters       // Parse using inclusiveSize
}

RibbonEmitter {
    u32         inclusiveSize
    Node        node
    f32         heightAbove
    f32         heightBelow
    f32         alpha
    float[3]    color               // RGB
    f32         lifespan
    u32         textureSlot
    u32         emissionRate
    u32         rows
    u32         columns
    u32         materialId          // Index into MTLS
    f32         gravity

    (KRHA)                          // Height above: f32
    (KRHB)                          // Height below: f32
    (KRAL)                          // Alpha: f32
    (KRCO)                          // Color: float[3]
    (KRTX)                          // Texture slot: u32
    (KRVS)                          // Visibility: f32
}
```

### 7.19 EVTS — Event Objects

**Tag:** `0x53545645` · **No inclusiveSize**

Event objects trigger sound effects, footprints, blood splats, and other environmental effects at specific animation frames.

```
EVTS {
    EventObject[?] objects
}

EventObject {
    Node        node
    KEVT        tracks              // Event timing data (see Section 8.3)
}
```

**Size computation:** Each EventObject is `node.inclusiveSize` + the KEVT block size. See Section 8.3 for the KEVT format.

### 7.20 CAMS — Cameras

**Tag:** `0x534D4143` · **Variable-sized objects**

```
CAMS {
    Camera[?] cameras               // Parse using inclusiveSize
}

Camera {
    u32         inclusiveSize
    char[80]    name
    float[3]    position            // Camera position (x, y, z)
    f32         fieldOfView         // Radians
    f32         farClippingPlane
    f32         nearClippingPlane
    float[3]    targetPosition      // Look-at target (x, y, z)

    (KCTR)                          // Position: float[3]
    (KCRL)                          // Rotation: f32 (scalar angle)
    (KTTR)                          // Target position: float[3]
}
```

### 7.21 CLID — Collision Shapes

**Tag:** `0x44494C43` · **No inclusiveSize**

```
CLID {
    CollisionShape[?] shapes
}

CollisionShape {
    Node        node
    u32         type                // See ShapeType enum

    float[vertexCount * 3] vertices // See vertex counts below

    // type == Sphere (2) or type == Cylinder (3):
    f32         radius
}
```

**ShapeType Enum and Vertex Counts**

| Value | Type | Vertex Count | Extra |
|---|---|---|---|
| 0 | Box | 2 | Two corners (min, max) |
| 1 | Plane | 2 | Two corners |
| 2 | Sphere | 1 | Center point + radius |
| 3 | Cylinder | 2 | Two endpoints + radius |

**Size computation:** Node `inclusiveSize` + 4 (type) + vertexCount × 12 (vertices) + optional 4 (radius if sphere or cylinder).

## 7.22 BPOS — Bind Poses

**Tag:** `0x534F5042` · **Version > 800**

```
BPOS {
    u32         count               // Number of bind pose matrices
    float[count * 12] matrices      // 3×4 row-major matrices (48 bytes
each)
}
```

Each matrix is a 3×4 affine transform (3 rows × 4 columns = 12 floats). There is one bind pose per node (including cameras), so the count typically equals the PIVT count + camera count.

## 7.23 FAFX — Face Effects

**Tag:** `0x58464146` · **Element size:** 340 bytes · **Count:** `size / 340` · **Version > 800**

```
FAFX {
    FaceEffect[size / 340] effects
}

FaceEffect {                        // 340 bytes
    char[80]    target              // Target name (e.g., "Node")
    char[260]   path                // FaceFX file path (.facefx)
}
```

## 7.24 CORN — Popcorn Emitters

**Tag:** `0x4E524F43` · **Variable-sized objects** · **Version > 800**

Emitters that use the PopcornFX particle runtime.

```
CORN {
    CornEmitter[?] emitters         // Parse using inclusiveSize
}

CornEmitter {
    u32         inclusiveSize
    Node        node
    f32         lifeSpan
    f32         emissionRate
    f32         speed
    float[4]    color               // RGBA (4 floats)
    u32         replaceableId
    char[260]   path                // Effect file path (.pkfx)
    char[260]   animVisibilityGuide // Sequence-based visibility rules
(see below)

    (KPPA)                          // Alpha: f32
    (KPPC)                          // Color: float[4] (RGBA)
    (KPPE)                          // Emission rate: f32
    (KPPL)                          // Lifespan: f32
    (KPPS)                          // Speed: f32
    (KPPV)                          // Visibility: f32
}
```

> **Important:** The KPPC color track uses `float[4]` (RGBA), not `float[3]` (RGB). This differs from GhostWolf's specification and from the KRCO ribbon emitter color track which uses `float[3]`.

### animVisibilityGuide

The `animVisibilityGuide` field is a comma-separated list of `Key=Value` rules that control when the PopcornFX emitter is active, based on the current animation sequence being played by the model. This provides a declarative way to tie particle emission to WC3's animation state machine without requiring per-frame KPPV visibility keyframes in every sequence.

**Format:** `SequenceName=State[, SequenceName=State, ...]`

The field is a null-terminated ASCII string within a 260-byte buffer. It may be empty (all zeroes), meaning the emitter is always active with no sequence-based toggling.

**Values (case-insensitive in practice):**

| Value | Meaning |
|---|---|
| `on` | Emitter is active during sequences matching this key |
| `off` | Emitter is suppressed during sequences matching this key |
| `On` / `Off` | Case variants observed in corpus (equivalent to lowercase) |
| `FStart/On-FEnd/Off` | Frame-range control: on at frame start, off at frame end (rare, 2 occurrences) |

**Key names** correspond to WC3 animation sequence names as defined in the SEQS chunk. The special key `Always` acts as a default rule. Common patterns:

| Pattern | Meaning | Example |
|---|---|---|
| `Always=on, Death=off` | On by default, off during death | Persistent auras, weapon glow |
| `Always=off, Death=on` | Off by default, on during death | Death explosions, blood splatter |
| `Always=off, Birth=on` | Off by default, on during birth | Building construction FX |
| `Always=off, Stand Work=on` | Off by default, on when building is working | Building work activity FX |
| `Always=on, Death=off, Decay=off` | On by default, off during death/decay | Common for spell effects |
| `Always=On, Death=Off, Dissipate=Off, Portrait=Off` | Hero glow: off during death/dissipate/portrait | Hero aura particles |
| `Always=off, Stand=on` | Off by default, on during idle | Brazier flames, ambient FX |
| `Always=off, Walk=on` | Off by default, on when walking | Dust trails |

**Corpus statistics (5,081 v1200 files):**

- **1,412 files** contain CORN chunks with **2,747 total emitters**
- **2,625** emitters (95.6%) have non-empty guides; **122** (4.4%) are empty
- **149 unique** guide strings observed; **96 unique** key names
- **5 value states:** `on` (2,569×), `off` (2,824×), `On` (222×), `Off` (647×), `FStart/On-FEnd/Off` (2×)

**Most frequent guide strings:**

| Count | Guide String |
|---|---|
| 430 | `Always=off, Death=on` |
| 321 | `Always=on, Death=off, Decay=off` |
| 286 | `Always=on, Death=off` |
| 225 | `Always=off, Birth=on` |
| 195 | `Always=On, Death=Off, Dissipate=Off, Portrait=Off` |
| 128 | `Always=off, Stand Work=on` |
| 122 | *(empty)* |
| 90 | `Always=off, Stand=on` |
| 79 | `Always=on` |
| 59 | `Always=off, Walk=on` |

**Key frequency (top 20):**

| Key | Count | Description |
|---|---|---|
| `Always` | 2,492 | Default state (on/off baseline) |
| `Death` | 1,491 | Death animation sequence |
| `Decay` | 380 | Corpse decay sequence |
| `Dissipate` | 333 | Hero/summoned unit dissipate |
| `Portrait` | 308 | Portrait/UI view mode |
| `Birth` | 264 | Construction/birth sequence |
| `Stand` | 150 | Idle standing animation |
| `Stand Work` | 148 | Building working (training/researching) |
| `Walk` | 64 | Walk animation |
| `Spell` | 45 | Spell cast animation |
| `Attack` | 28 | Attack animation |
| `Death Alternate` | 27 | Alternate death animation |
| `Stand Channel` | 26 | Channeling spell idle |
| `Death 1` | 22 | Death variation 1 |
| `Morph` | 22 | Morph/transform animation |
| `Spell Slam` | 18 | Slam spell animation |
| `Spell 1` | 16 | Spell variation 1 |
| `Morph Alternate` | 16 | Alternate morph animation |
| `Stand Work Gold` | 14 | Gold-gathering work animation |
| `Walk 1` | 14 | Walk variation 1 |

> **Note:** Some corpus entries contain typos (`Alwyas`, `Allways`, `deacy` for `Decay`) and inconsistent casing (`always` vs `Always`, `Off` vs `off`). The engine likely performs case-insensitive matching. Some entries have trailing `\r\n` sequences or extra whitespace, indicating imprecise editor tooling.

---

# 8. Animation Tracks

## 8.1 Track Structure

All animation tracks (except KEVT) follow a uniform structure. The track tag identifies both the parent object and the data type of keyframes.

```
TrackChunk {
    FourCC      tag                 // Track identifier (e.g., KGTR, KMTA,
etc.)
    u32         keyCount            // Number of keyframes
    u32         interpolationType   // See InterpolationType enum
    u32         globalSequenceId    // Index into GLBS, or 0xFFFFFFFF (-1)
for none

    Keyframe[keyCount] keyframes
}
```

**Keyframe**

```
Keyframe {
    i32         frame               // Time in milliseconds (may be
negative in some models)
    T           value               // Keyframe value (type depends on
track tag)

    // interpolationType > 1 (Hermite or Bezier):
    T           inTan               // In-tangent
    T           outTan              // Out-tangent
}
```

**InterpolationType Enum**

| Value | Type | Description |
|---|---|---|
| 0 | None | No interpolation (stepped) |
| 1 | Linear | Linear interpolation |
| 2 | Hermite | Hermite spline (uses inTan/outTan) |
| 3 | Bezier | Bezier spline (uses inTan/outTan) |

## 8.2 Track Tag Reference

The following table lists all track tags, their data type `T`, parent object, and semantic meaning.

### Node Tracks

| Tag | Data Type | Description |
|---|---|---|
| **KGTR** | `float[3]` | Translation (x, y, z) |
| **KGRT** | `float[4]` | Rotation (quaternion: x, y, z, w) |
| **KGSC** | `float[3]` | Scaling (x, y, z) |

### Layer Tracks

| Tag | Data Type | Description | Version |
|---|---|---|---|
| **KMTF** | `u32` | Texture ID | All (per-SubTexture in ≥ 1100) |
| **KMTA** | `f32` | Alpha | All |
| **KMTE** | `f32` | Emissive gain | > 800 |
| **KFC3** | `float[3]` | Fresnel color (RGB) | > 900 |
| **KFCA** | `f32` | Fresnel alpha | > 900 |
| **KFTC** | `f32` | Fresnel team color | > 900 |

### Texture Animation Tracks

| Tag | Data Type | Description |
|---|---|---|
| **KTAT** | `float[3]` | Translation (u, v, w) |
| **KTAR** | `float[4]` | Rotation (quaternion XYZW) |
| **KTAS** | `float[3]` | Scaling (u, v, w) |

> **Note:** KTAR uses a quaternion `float[4]`, same as node rotation KGRT.

### Geoset Animation Tracks

| Tag | Data Type | Description |
|---|---|---|
| **KGAO** | `f32` | Alpha |
| **KGAC** | `float[3]` | Color (BGR) |

### Light Tracks

| Tag | Data Type | Description |
|---|---|---|
| **KLAS** | `f32` | Attenuation start |
| **KLAE** | `f32` | Attenuation end |
| **KLAC** | `float[3]` | Color (RGB) |
| **KLAI** | `f32` | Intensity |
| **KLBI** | `f32` | Ambient intensity |
| **KLBC** | `float[3]` | Ambient color (RGB) |
| **KLAV** | `f32` | Visibility |

### Attachment Tracks

| Tag | Data Type | Description |
|---|---|---|
| **KATV** | `f32` | Visibility |

### Particle Emitter Tracks (PREM)

| Tag | Data Type | Description |
|---|---|---|
| **KPEE** | `f32` | Emission rate |
| **KPEG** | `f32` | Gravity |
| **KPLN** | `f32` | Longitude |
| **KPLT** | `f32` | Latitude |
| **KPEL** | `f32` | Lifespan |
| **KPES** | `f32` | Speed |
| **KPEV** | `f32` | Visibility |

### Particle Emitter 2 Tracks (PRE2)

| Tag | Data Type | Description |
|---|---|---|
| **KP2S** | `f32` | Speed |
| **KP2R** | `f32` | Variation |
| **KP2L** | `f32` | Latitude |
| **KP2G** | `f32` | Gravity |
| **KP2E** | `f32` | Emission rate |
| **KP2N** | `f32` | Length |
| **KP2W** | `f32` | Width |
| **KP2V** | `f32` | Visibility |

### Ribbon Emitter Tracks

| Tag | Data Type | Description |
|---|---|---|
| **KRHA** | `f32` | Height above |
| **KRHB** | `f32` | Height below |
| **KRAL** | `f32` | Alpha |
| **KRCO** | `float[3]` | Color (RGB) |
| **KRTX** | `u32` | Texture slot |
| **KRVS** | `f32` | Visibility |

### Camera Tracks

| Tag | Data Type | Description |
|---|---|---|
| **KCTR** | `float[3]` | Position (x, y, z) |
| **KCRL** | `f32` | Target rotation (scalar angle) |
| **KTTR** | `float[3]` | Target position (x, y, z) |

**Popcorn Emitter Tracks (CORN)**

| Tag | Data Type | Description |
|---|---|---|
| **KPPA** | `f32` | Alpha |
| **KPPC** | `float[4]` | Color (RGBA) |
| **KPPE** | `f32` | Emission rate |
| **KPPL** | `f32` | Lifespan |
| **KPPS** | `f32` | Speed |
| **KPPV** | `f32` | Visibility |

**Sound Emitter Tracks (SNEM)**

| Tag | Data Type | Description |
|---|---|---|
| **KSEK** | `u32` | Sound event track |

> **Note:** KSEK follows the standard track structure (with interpolation type and global sequence ID in the header), unlike KEVT which has a unique layout.

### 8.3 KEVT — Event Tracks

Event object tracks have a unique structure that differs from all other track types. They store only frame times with no interpolation or value data — there is no `interpolationType` field.

```
KEVT {
    FourCC      "KEVT"              // 0x5456454B
    u32         trackCount          // Number of event times
    u32         globalSequenceId    // Index into GLBS, or 0xFFFFFFFF (-1)
    i32[trackCount] tracks          // Frame times (ms) when the event
fires
}
```

> **Field order:** `globalSequenceId` sits **between** `trackCount` and the track data — the same position it occupies in a standard track header, just with the `interpolationType` field omitted. (Verified against war3.w3mod game assets and the mdx-m3-viewer reference reader.) The track times are signed `i32`, consistent with keyframe `frame` values elsewhere.

---

# 9. Node Structure

The `Node` is a shared structure embedded in Bones, Lights, Helpers, Attachments, Particle Emitters, Ribbon Emitters, Event Objects, Collision Shapes, and Popcorn Emitters. It defines the object's place in the scene hierarchy and its animation channels.

```
Node {
    u32         inclusiveSize       // Total size of this Node including
this field
    char[80]    name                // Node name
    u32         objectId            // Unique node ID
    u32         parentId            // Parent node ID, or 0xFFFFFFFF (-1)
for root
    u32         flags               // See NodeFlags bitfield

    (KGTR)                          // Translation: float[3]
    (KGRT)                          // Rotation: float[4] (quaternion)
    (KGSC)                          // Scaling: float[3]
}
```

## NodeFlags Bitfield

The `flags` field encodes both the node type **and** behavioral flags.

### Node Type Flags

| Bit | Hex | Type |
|---|---|---|
| 8 | 0x100 | Bone |
| 9 | 0x200 | Light |
| 10 | 0x400 | Event object |
| 11 | 0x800 | Attachment |
| 12 | 0x1000 | Particle emitter |
| 13 | 0x2000 | Collision shape |
| 14 | 0x4000 | Ribbon emitter |
| 17 | 0x20000 | Line emitter / Particle emitter 2 |

If none of the type bits are set (`flags & 0xFF00 == 0`), the node is a Helper.

**Behavioral Flags**

| Bit | Hex | Description |
|---|---|---|
| 0 | 0x1 | Don't inherit translation |
| 1 | 0x2 | Don't inherit rotation |
| 2 | 0x4 | Don't inherit scaling |
| 3 | 0x8 | Billboarded |
| 4 | 0x10 | Billboarded lock X |
| 5 | 0x20 | Billboarded lock Y |
| 6 | 0x40 | Billboarded lock Z |
| 7 | 0x80 | Camera anchored |
| 15 | 0x8000 | Emitter uses MDL (PREM) / Unshaded (PRE2) |
| 16 | 0x10000 | Emitter uses TGA (PREM) / Sort primitives far Z (PRE2) |
| 18 | 0x40000 | Unfogged |
| 19 | 0x80000 | Model space |
| 20 | 0x100000 | XY quad |

> **Note on multiple roots:** MDX models commonly have multiple root nodes (parentId = 0xFFFFFFFF). For example, unit models often have separate bone hierarchies for the living model and its corpse/decay animation.

---

# 10. Version Differences

This section summarizes the structural changes introduced in each version.

## Version 800 (Classic)

The base format. All structures documented above without version conditions apply.

## Version 900 (Reforged Beta)

Added to existing structures:

- **Material**: `char[80] shader` field after `flags`
- **Layer**: `emissiveGain` field after `alpha`; KMTE track
- **Geoset**: `lod` (`u32`) and `lodName` (`char[80]`) after `selectionFlags`; optional TANG and SKIN sub-chunks

New chunks:

- **BPOS**: Bind pose matrices
- **FAFX**: Face animation effects (FaceFX)
- **CORN**: Popcorn particle emitters

## Version 1000 (Reforged — First Release)

Added to existing structures:

- **Layer**: `fresnelColor[3]`, `fresnelOpacity`, `fresnelTeamColor` fields after `emissiveGain`; KFC3, KFCA, KFTC Fresnel animation tracks

> The fresnel **fields** and fresnel **tracks** were added together in v1000. A v900 layer has `emissiveGain` but no fresnel data at all.

## Version 1100 (Reforged — 2.0.0)

Major layer system change:

- **Material**: `shader` string field **removed** (was only for 800 < version < 1100)
- **Layer**: SubTexture system added — `shader` (`ShaderType u32`), `numSubTextures`, and per-subtexture `{textureId, slot, KMTF}`
- **Layer**: KMTF track moved from Layer level to per-SubTexture

## Version 1200 (Reforged Current)

Added to existing structures:

- **Light**: `shadowIntensity` field (`f32`) after `ambientIntensity`

---

# 11. Implementation Notes

## Byte Order

All multi-byte values are **little-endian**.

## String Fields

Fixed-length string fields (`char[80]`, `char[260]`) are null-padded. The actual string content ends at the first null byte; remaining bytes should be zero. When writing, pad with null bytes to the full field length.

## Animation Timing

Warcraft III animations use millisecond timing. At 60 FPS, advance animation counters by `1000/60 ≈ 16.67` ms per frame. Keyframe times in tracks and sequence intervals are in milliseconds.

## Rotations

- **Node rotations (KGRT):** Quaternions as `float[4]` in (x, y, z, w) order.
- **Texture animation rotations (KTAR):** Quaternions as `float[4]` in (x, y, z, w) order.
- **Camera rotations (KCRL):** Scalar `f32` angle.

## Inclusive Size Fields

The `inclusiveSize` at the start of variable-sized objects includes the 4 bytes of the `inclusiveSize` field itself. For example, if `inclusiveSize = 100`, the object occupies exactly 100 bytes from the start of the `inclusiveSize` field (96 bytes of payload after it).

## Geoset Sub-Chunk Tag-Size Pairs

Within a Geoset, sub-chunks like VRTX, NRMS, PTYP, etc. use a tag + count pattern (not tag + byte-size). The count represents the number of elements, not bytes. For example:

- VRTX count = number of vertices (each vertex is 3 floats = 12 bytes)
- PVTX count = number of face indices (each is 1 `u16` = 2 bytes)
- GNDX count = number of vertex groups (each is 1 `u8` = 1 byte)

## Global Sequence IDs

A `globalSequenceId` of `0xFFFFFFFF` (-1 as signed) means the track uses normal sequence-based animation rather than a global sequence loop.

## Parsing Tracks Within Objects

To determine whether optional animation tracks exist within an object, either:

1. **Size tracking:** Track bytes consumed vs. `inclusiveSize` — if bytes remain, the next 4 bytes may be a track tag.
2. **Tag peeking:** Read the next 4 bytes and check against known track tags for that object type. If it matches, parse it; otherwise, stop.

Approach 1 is more robust as it handles unknown track types gracefully.

### Chunk Tag Constants

For reference, here are the FourCC tag values as `u32` little-endian:

```
MDLX = 0x584C444D      VERS = 0x53524556      MODL = 0x4C444F4D
SEQS = 0x53514553      GLBS = 0x53424C47      TEXS = 0x53584554
SNDS = 0x53444E53      SNEM = 0x4D454E53      MTLS = 0x534C544D      TXAN
= 0x4E415854
GEOS = 0x534F4547      GEOA = 0x414F4547      BONE = 0x454E4F42
LITE = 0x4554494C      HELP = 0x504C4548      ATCH = 0x48435441
PIVT = 0x54564950      PREM = 0x4D455250      PRE2 = 0x32455250
RIBB = 0x42424952      EVTS = 0x53545645      CAMS = 0x534D4143
CLID = 0x44494C43      BPOS = 0x534F5042      FAFX = 0x58464146
CORN = 0x4E524F43

LAYS = 0x5359414C      VRTX = 0x58545256      NRMS = 0x534D524E
PTYP = 0x50595450      PCNT = 0x544E4350      PVTX = 0x58545650
GNDX = 0x58444E47      MTGC = 0x4347544D      MATS = 0x5354414D
TANG = 0x474E4154      SKIN = 0x4E494B53      UVAS = 0x53415655
UVBS = 0x53425655      KEVT = 0x5456454B      KSEK = 0x4B45534B
```

## Track Tag Constants

```
KGTR = 0x5254474B      KGRT = 0x5452474B      KGSC = 0x4353474B
KMTF = 0x46544D4B      KMTA = 0x41544D4B      KMTE = 0x45544D4B
KFC3 = 0x3343464B      KFCA = 0x4143464B      KFTC = 0x4354464B
KTAT = 0x5441544B      KTAR = 0x5241544B      KTAS = 0x5341544B
KGAO = 0x4F41474B      KGAC = 0x4341474B
KLAS = 0x53414C4B      KLAE = 0x45414C4B      KLAC = 0x43414C4B
KLAI = 0x49414C4B      KLBI = 0x49424C4B      KLBC = 0x43424C4B
KLAV = 0x56414C4B
KATV = 0x5654414B
KPEE = 0x4545504B      KPEG = 0x4745504B      KPLN = 0x4E4C504B
KPLT = 0x544C504B      KPEL = 0x4C45504B      KPES = 0x5345504B
KPEV = 0x5645504B
KP2S = 0x5332504B      KP2R = 0x5232504B      KP2L = 0x4C32504B
KP2G = 0x4732504B      KP2E = 0x4532504B      KP2N = 0x4E32504B
KP2W = 0x5732504B      KP2V = 0x5632504B
KRHA = 0x4148524B      KRHB = 0x4248524B      KRAL = 0x4C41524B
KRCO = 0x4F43524B      KRTX = 0x5854524B      KRVS = 0x5356524B
KCTR = 0x5254434B      KCRL = 0x4C52434B      KTTR = 0x5254544B
KPPA = 0x4150504B      KPPC = 0x4350504B      KPPE = 0x4550504B
KPPL = 0x4C50504B      KPPS = 0x5350504B      KPPV = 0x5650504B
KSEK = 0x4B45534B
```

---

# Appendix A — PopcornFX Integration

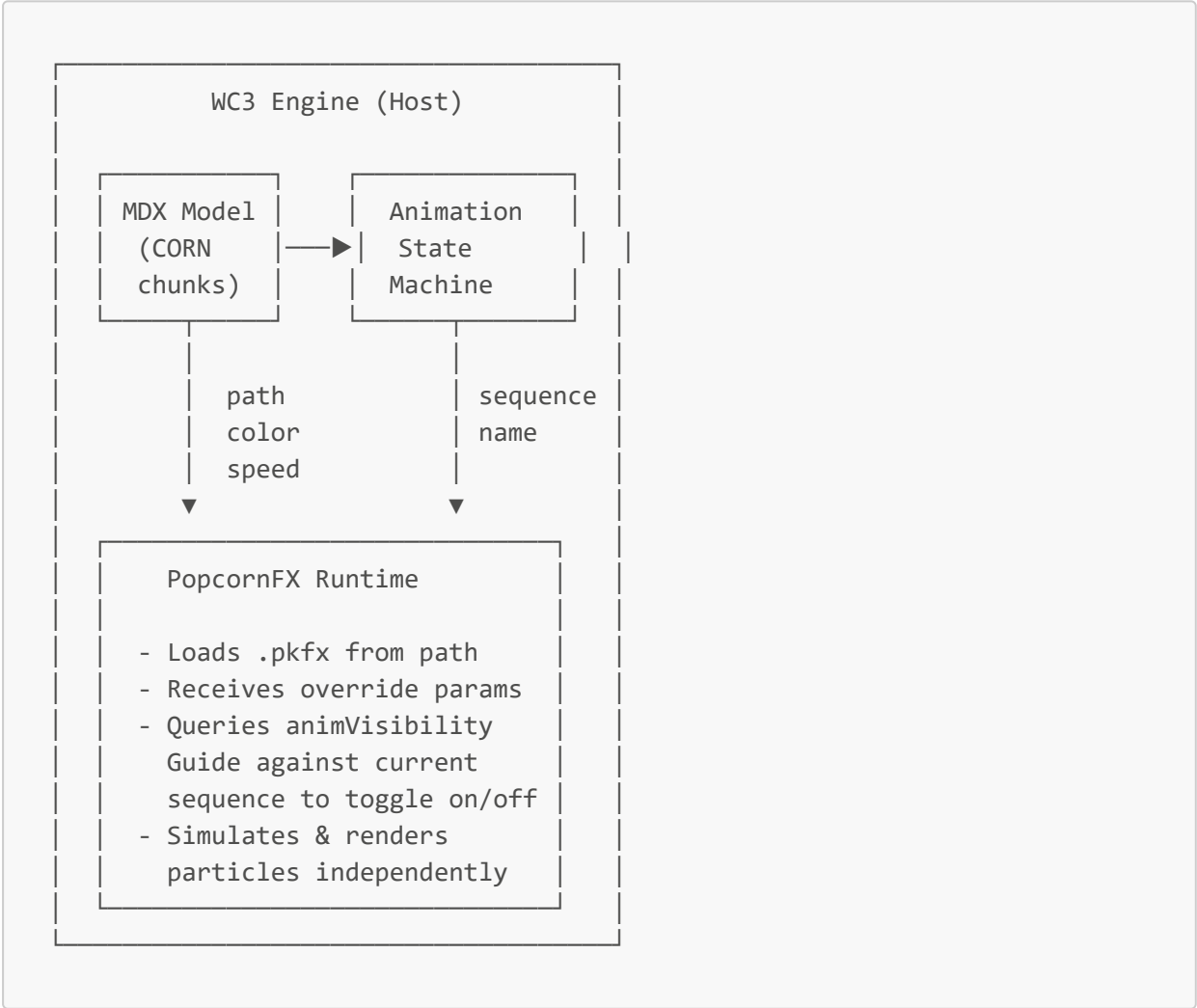
## Overview

Warcraft III: Reforged (version > 800) replaced the classic particle system with **PopcornFX**, a commercial real-time particle effects middleware by Persistant Studios. Instead of the engine computing particle physics from inline parameters (as PREM and PRE2 do), CORN emitters delegate particle simulation entirely to the PopcornFX runtime, which loads external effect definition files.

This is a fundamental architectural shift: classic emitters (PREM, PRE2) store all particle behavior inside the MDX binary — spawn rates, physics, textures, color curves — while CORN emitters are lightweight bridge objects that point to an external `.pkfx` effect file and pass a small set of override parameters.

## Runtime Architecture

The integration follows a host-plugin model:



1. The engine loads the MDX model and encounters one or more `CornEmitter` objects in the CORN chunk.
2. Each emitter's `Node` places it in the scene hierarchy (parented to bones, etc.), giving it a world-space transform.
3. The engine passes the emitter's `path` to the PopcornFX runtime, which loads the `.pkfx` effect definition.
4. Per-frame, the engine feeds the PopcornFX runtime the emitter's current transform (from the Node hierarchy), and the override parameters: `color`, `speed`, `emissionRate`, `lifeSpan` — including any animated values from KPPC/KPPS/KPPE/KPPL tracks.
5. The engine evaluates the `animVisibilityGuide` string against the currently playing animation sequence name. If the guide maps the current sequence to `off`, the emitter is suppressed; if `on`, it is active. The KPPV visibility track provides additional per-keyframe control.
6. The PopcornFX runtime handles all particle spawning, simulation, and rendering autonomously.

### .pkfx Files

PopcornFX effect files (`.pkfx`) are **baked binary effect packages** produced by the PopcornFX Editor. They are not part of the MDX format — they are external assets loaded at runtime by the PopcornFX plugin.

A `.pkfx` file encapsulates:

| Component | Description |
|---|---|
| **Particle layers** | One or more particle layers, each defining spawn shape, velocity, physics, and lifetime |
| **Renderers** | Billboard, ribbon, mesh, or light renderers with material/blend settings |
| **Textures & atlases** | Texture references (resolved relative to the game's asset root), sprite sheet definitions |
| **Evolvers** | Gravity, turbulence, collisions, attractors, and other physics modules |
| **Samplers** | Curve samplers for color-over-life, size-over-life, etc. |
| **Events & triggers** | Spawn-on-death, sub-effect triggers, sound hooks |
| **Spatial layers** | LOD settings, bounding volumes for culling |

The MDX `CornEmitter` does **not** replicate any of this data — it only stores the path to the `.pkfx` file and a small set of host-side overrides.

> **Note:** The `.pkfx` format is proprietary to PopcornFX and is not documented here. The PopcornFX SDK documentation covers the baked effect format internals.

### Path Conventions

The `path` field in `CornEmitter` is a 260-byte null-terminated string containing a relative path to the `.pkfx` asset. Paths use forward slashes (`/`) as separators, though some corpus entries use backslashes (`\`). The path is relative to the game's asset root directory.

**Directory structure observed in the corpus (2,747 emitters, 766 unique paths):**

| Top-Level Directory | Count | Description |
|---|---|---|
| `SharedFX/` | 1,597 | Reusable effects shared across many models |
| `Abilities/` | 335 | Spell and ability effects |
| `Units/` | 312 | Unit-specific effects (auras, weapon glow, etc.) |
| `Buildings/` | 274 | Building construction, work, and destruction FX |
| `Doodads/` | 89 | Doodad/prop effects and cinematic FX |
| `Objects/` | 69 | Item and object effects |
| `Environment/` | 24 | Environmental/weather effects |
| `UI/` | 1 | UI overlay effects |

Typical path structure: `{Category}/{Subcategory}/{EffectName}/{EffectName}.pkfx`

**Examples:**

```
SharedFX/HUBuildingDeath_Small/HUBuildingDeath_Small.pkfx   (266 emitters)
SharedFX/Hero_Glow/Hero_Glow.pkfx                          (196 emitters)
SharedFX/HumanBuildingBirth_Small/HumanBuildingBirth_Small.pkfx  (185
emitters)
Units/Demon/Infernal/Infernal.pkfx                          (20 emitters)
Abilities/Spells/Undead/FrostNova/FrostNovaTarget.pkfx      (spell FX)
Buildings/Human/AltarOfKings/AOK_Glow.pkfx                 (building FX)
```

The `SharedFX/` directory is heavily reused — a single effect like `HUBuildingDeath_Small.pkfx` is referenced by 266 different emitters across many building models, enabling a unified destruction look.

> **Corpus note:** 26 paths (< 1%) lack the `.pkfx` extension (e.g., `SharedFX/Ship_Wake_Small/`), likely authoring errors that the engine resolves by appending a default extension or searching the directory. 7 paths contain absolute developer paths (`D:\Warcraft3\warcraft3\branches\v1.32.0.5\...`), leaked from development builds.

### MDX-to-PopcornFX Data Flow

The `CornEmitter` fields map to the PopcornFX runtime as follows:

| MDX Field | PopcornFX Role | Notes |
|---|---|---|
| `path` | Effect file to load | Resolved against asset root |
| `color` (float[4] RGBA) | Tint override | Passed to effect as an attribute sampler; many effects use black (0,0,0,1) as neutral |
| `speed` | Speed multiplier | Scales particle velocity in the effect |
| `emissionRate` | Emission rate multiplier | Scales spawn count |
| `lifeSpan` | Lifetime multiplier | Scales particle lifetime |
| `replaceableId` | Texture replacement ID | For team-color or replaceable texture effects (0 = none) |
| `Node` transform | World-space transform | The emitter inherits position/rotation/scale from the bone hierarchy |
| KPPA track | Animated alpha | Per-keyframe alpha override |
| KPPC track | Animated color | Per-keyframe RGBA color override |
| KPPE track | Animated emission rate | Per-keyframe emission rate override |
| KPPL track | Animated lifespan | Per-keyframe lifespan override |
| KPPS track | Animated speed | Per-keyframe speed override |
| KPPV track | Animated visibility | Per-keyframe visibility (0.0 = hidden, 1.0 = visible) |

The static fields provide default values. When animation tracks are present, the tracked values override the static defaults at each keyframe.

### Visibility Control

CORN emitters have a **two-layer visibility system** that is unique among MDX object types:

**Layer 1 — `animVisibilityGuide` (sequence-level):** A declarative string that maps animation sequence names to on/off states. The engine evaluates this against the model's current sequence (from SEQS) to determine whether the emitter should be active at all. See Section 7.24 for the full guide format, key reference, and corpus statistics.

**Layer 2 — KPPV track (keyframe-level):** A standard `f32` animation track that provides per-keyframe visibility control within a sequence. Values of `0.0` suppress the emitter; `1.0` makes it

visible.

The two layers combine: the `animVisibilityGuide` acts as a coarse gate (is this emitter relevant for the current animation?), while KPPV provides fine-grained per-frame control within active sequences.

This dual system lets artists achieve common patterns efficiently:

- **"Always on except during death"** — Set `Always=on, Death=off` in the guide instead of keyframing KPPV across every sequence.
- **"Only during spell cast, with fade-in"** — Set `Always=off, Spell=on` in the guide, then use a KPPV hermite track for the fade-in curve within the Spell sequence.

## Corpus Statistics

| Metric | Value |
|---|---|
| Files with CORN chunks | 1,412 / 5,081 (27.8%) |
| Total CORN emitters | 2,747 |
| Unique .pkfx paths | 766 |
| Emitters with non-empty guide | 2,625 (95.6%) |
| Emitters with empty guide | 122 (4.4%) |
| Unique guide strings | 149 |
| Unique guide key names | 96 |
| Most-referenced effect | `SharedFX/HUBuildingDeath_Small/HUBuildingDeath_Small.pkfx` (266×) |

# Credits & References

- **WhiteoutLib** — C++ parser and writer implementation used as the primary reference for this specification
- **GhostWolf's MDX Specifications** — HiveWorkshop thread (2013–2019), the original community specification
- **Barncastle's MDXReForged** — C# implementation documenting Reforged-era changes (2019)
- **BlinkBoy** — Contributions on collision shape types, KEVT format, SND chunk, and various corrections
- **Magos' MDX Specifications** — Original reverse-engineering work that started community documentation
- **PopcornFX** — persistant-studios.com — Commercial particle effects middleware by Persistant Studios, integrated into WC3 Reforged

## Key Corrections vs. Prior Specifications

| Item | Old Spec | Corrected |
|---|---|---|
| KTAR data type | `f32` (scalar angle) | `float[4]` (quaternion XYZW) |
| KPPC data type | `float[3]` (RGB) | `float[4]` (RGBA) |
| KEVT field order | (v2.0 of this doc placed `globalSequenceId` **after** the track times) | `globalSequenceId` sits **between** `trackCount` and the track times — verified against game assets and mdx-m3-viewer |
| Material shader | Present when `version > 800` | Present when `version > 800 && version < 1100` |
| Light shadowIntensity | Not documented | Present when `version >= 1200` |
| Layer SubTexture system | Not documented | New in version ≥ 1100 |
| Layer fresnel fields | Present when `version > 800` (with `emissiveGain`) | `emissiveGain` is `version > 800`; `fresnelColor`/`fresnelOpacity`/`fresnelTeamColor` are `version > 900` |
| Layer v≥1100 field | `u32 isHD` (boolean) | `u32 shader` — a `ShaderType` enum (0 = SD, 1 = HD, …) |
| Keyframe `frame` | Unsigned time | Signed `i32`; negative frames occur in shipped models |